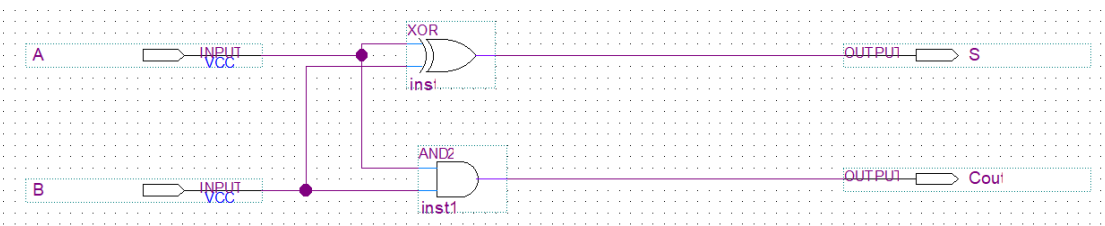


Accumulator

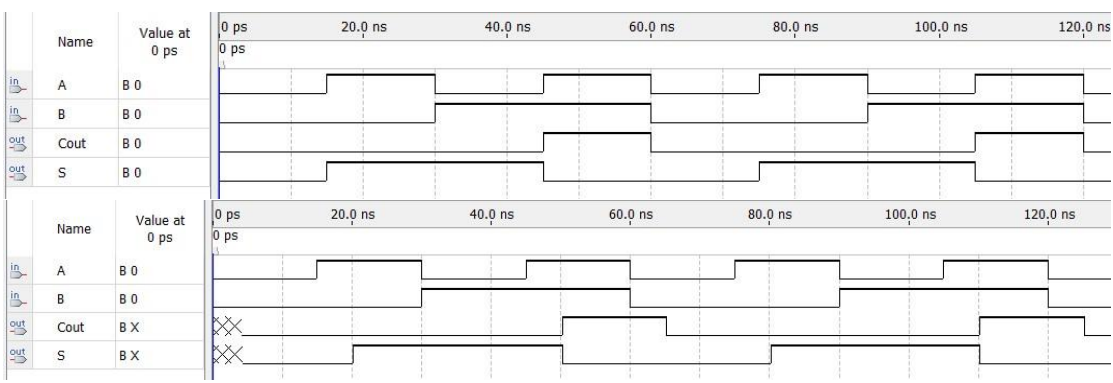
Before designing the Accumulator, a HalfAdder and a FullAdder design is needed, to make an 8-bit-Adder.

HalfAdder

Circuit:



The HalfAdder functions correctly, as it can be seen below:

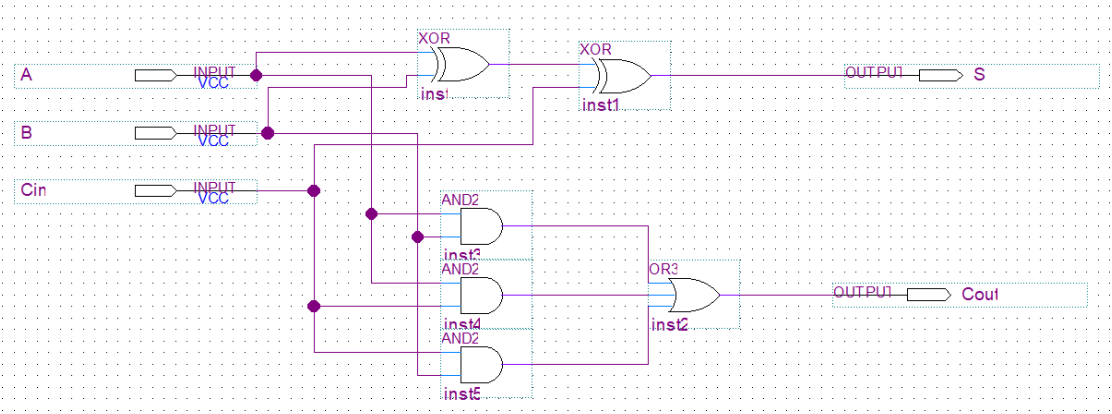


Static Analysis:

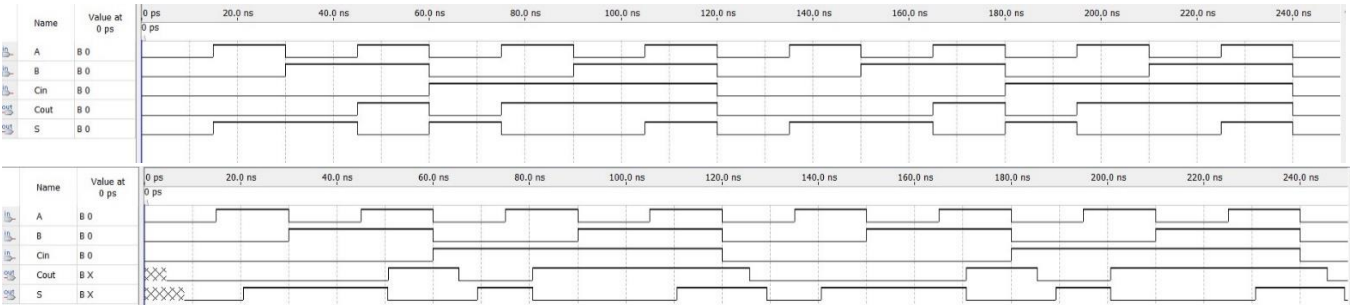
	Input Port	Output Port	RR	RF	FR	FF
1	A	Cout	5.258			5.258
2	A	S	5.256	5.256	5.256	5.256
3	B	Cout	5.431			5.431
4	B	S	5.431	5.431	5.431	5.431

FullAddder

Circuit:



The FullAddder functions correctly, as it can be seen below:

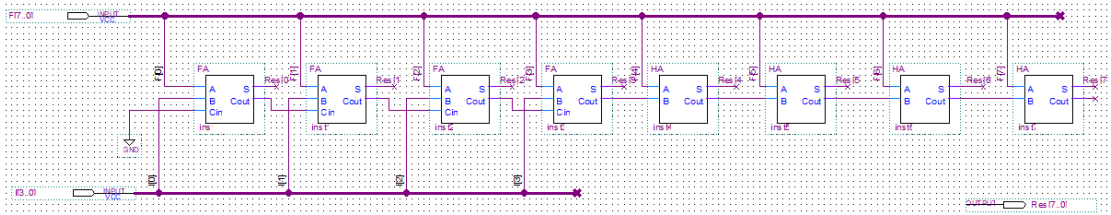


Static Analysis:

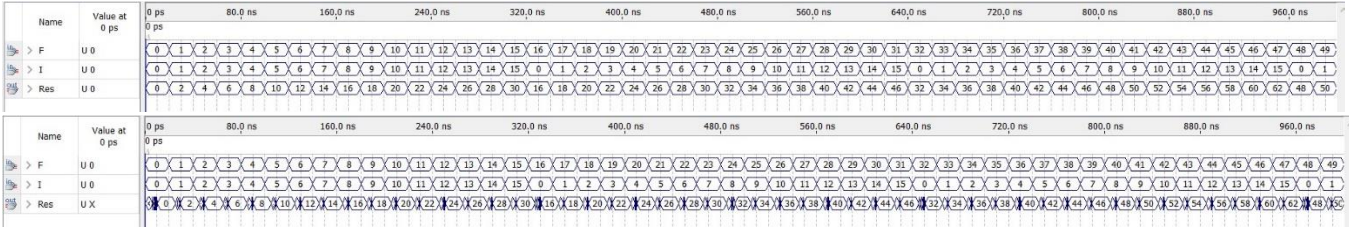
	Input Port	Output Port	RR	RF	FR	FF
1	A	Cout	5.667			5.667
2	A	S	5.655	5.655	5.655	5.655
3	B	Cout	5.358			5.358
4	B	S	5.347	5.347	5.347	5.347
5	Cin	Cout	9.282			9.282
6	Cin	S	9.276	9.276	9.276	9.276

8-Bit-Adder

Circuit:



The 8-bit-Adder functions correctly, as it can be seen below:



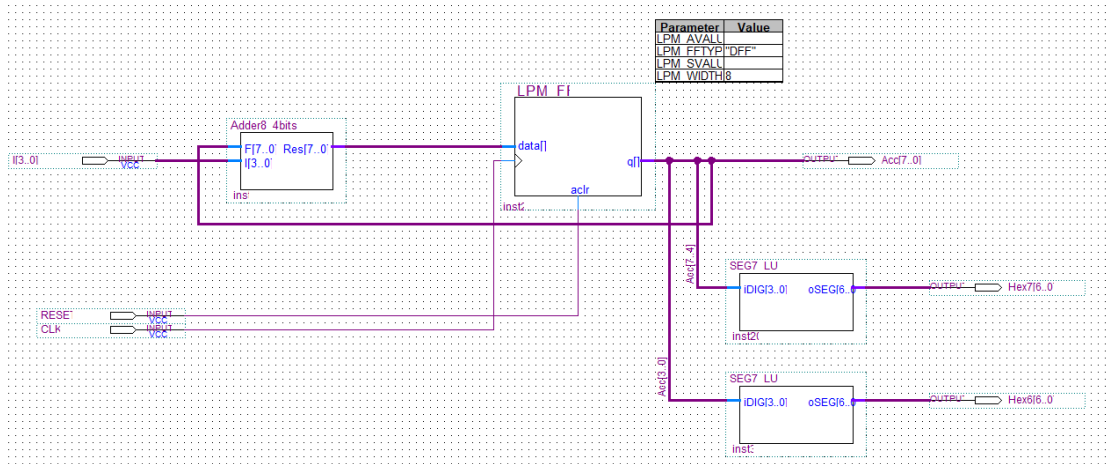
Static Analysis:

	Input Port	Output Port	RR	RF	FR	FF
1	F[0]	Res[0]	8.836	8.836	8.836	8.836
2	F[0]	Res[1]	9.107	9.107	9.107	9.107
3	F[0]	Res[2]	9.638	9.638	9.638	9.638
4	F[0]	Res[3]	10.029	10.029	10.029	10.029
5	F[0]	Res[4]	10.046	10.046	10.046	10.046
6	F[0]	Res[5]	12.121	12.121	12.121	12.121
7	F[0]	Res[6]	10.474	10.474	10.474	10.474
8	F[0]	Res[7]	11.903	11.903	11.903	11.903
9	F[1]	Res[1]	9.141	9.141	9.141	9.141
10	F[1]	Res[2]	9.670	9.670	9.670	9.670
11	F[1]	Res[3]	10.061	10.061	10.061	10.061
12	F[1]	Res[4]	10.078	10.078	10.078	10.078
13	F[1]	Res[5]	12.153	12.153	12.153	12.153
14	F[1]	Res[6]	10.506	10.506	10.506	10.506
15	F[1]	Res[7]	11.935	11.935	11.935	11.935
16	F[2]	Res[2]	9.424	9.424	9.424	9.424
17	F[2]	Res[3]	9.816	9.816	9.816	9.816
18	F[2]	Res[4]	9.833	9.833	9.833	9.833
19	F[2]	Res[5]	11.908	11.908	11.908	11.908
20	F[2]	Res[6]	10.261	10.261	10.261	10.261
21	F[2]	Res[7]	11.690	11.690	11.690	11.690
22	F[3]	Res[3]	9.482	9.482	9.482	9.482
23	F[3]	Res[4]	9.774	9.774	9.774	9.774
24	F[3]	Res[5]	11.846	11.846	11.846	11.846
25	F[3]	Res[6]	10.199	10.199	10.199	10.199
26	F[3]	Res[7]	11.628	11.628	11.628	11.628
27	F[4]	Res[4]	10.308	10.308	10.308	10.308
28	F[4]	Res[5]	12.379	12.379	12.379	12.379
29	F[4]	Res[6]	10.732	10.732	10.732	10.732
30	F[4]	Res[7]	12.161	12.161	12.161	12.161
31	F[5]	Res[5]	10.859	10.859	10.859	10.859
32	F[5]	Res[6]	9.211	9.211	9.211	9.211
33	F[5]	Res[7]	11.187	11.187	11.187	11.187
34	F[6]	Res[6]	6.385	6.385	6.385	6.385
35	F[6]	Res[7]	8.089	8.089	8.089	8.089
36	F[7]	Res[7]	8.173	8.173	8.173	8.173
37	I[0]	Res[0]	5.054	5.054	5.054	5.054
38	I[0]	Res[1]	5.323	5.323	5.323	5.323
39	I[0]	Res[2]	5.853	5.853	5.853	5.853
40	I[0]	Res[3]	6.244	6.244	6.244	6.244
41	I[0]	Res[4]	6.261	6.261	6.261	6.261
42	I[0]	Res[5]	8.336	8.336	8.336	8.336
43	I[0]	Res[6]	6.689	6.689	6.689	6.689
44	I[0]	Res[7]	8.118	8.118	8.118	8.118

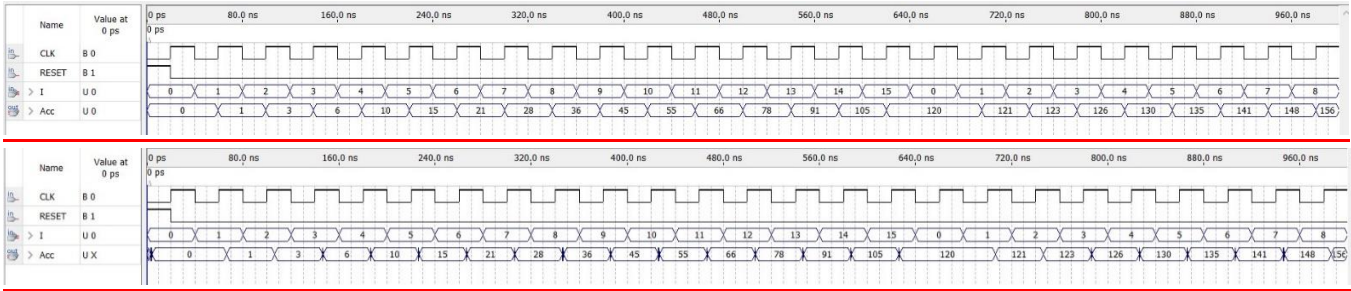
45	I[1]	Res[1]	5.436	5.436	5.436	5.436
46	I[1]	Res[2]	5.970	5.970	5.970	5.970
47	I[1]	Res[3]	6.361	6.361	6.361	6.361
48	I[1]	Res[4]	6.378	6.378	6.378	6.378
49	I[1]	Res[5]	8.453	8.453	8.453	8.453
50	I[1]	Res[6]	6.806	6.806	6.806	6.806
51	I[1]	Res[7]	8.235	8.235	8.235	8.235
52	I[2]	Res[2]	5.480	5.480	5.480	5.480
53	I[2]	Res[3]	5.870	5.870	5.870	5.870
54	I[2]	Res[4]	5.887	5.887	5.887	5.887
55	I[2]	Res[5]	7.962	7.962	7.962	7.962
56	I[2]	Res[6]	6.315	6.315	6.315	6.315
57	I[2]	Res[7]	7.744	7.744	7.744	7.744
58	I[3]	Res[3]	6.709	6.709	6.709	6.709
59	I[3]	Res[4]	6.998	6.998	6.998	6.998
60	I[3]	Res[5]	9.074	9.074	9.074	9.074
61	I[3]	Res[6]	7.427	7.427	7.427	7.427
62	I[3]	Res[7]	8.856	8.856	8.856	8.856

Accumulator

Circuit:



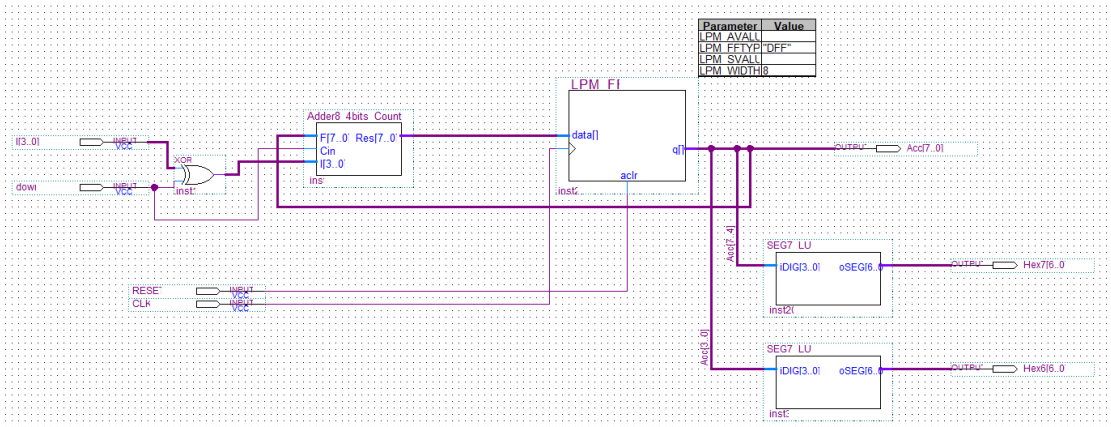
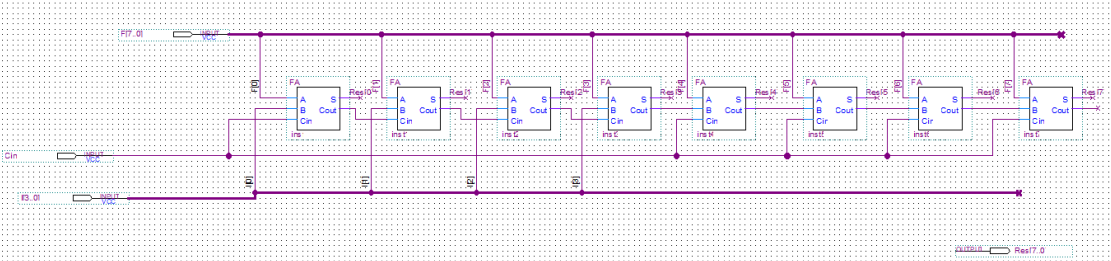
The Accumulator functions correctly, as it can be seen below:



Modded Accumulator

The following Accumulator is designed to increase/ decrease the output by a fixed step, which is defined by the input:

The new modded design can be seen below:



The Modded Accumulator functions correctly, as it can be seen below:

